

Trainings ▾

Fuel Recommendations

WARNING

Do not mix gasoline, alcohol, or gasohol with diesel fuel. This mixture can cause an explosion.

CAUTION

Because of the precise tolerances of diesel injection systems, fuel must be kept clean and free of dirt or water. Dirt or water in the system can cause severe damage to both the fuel pump and the injectors.

Cummins Inc. recommends the use of ASTM number 2D fuel. The use of number 2 diesel fuel will result in optimum engine performance.

At operating temperatures below 0°C [32°F], acceptable performance can be obtained by using blends of number 2D and number 1D.

Note : Lighter fuels can reduce fuel economy.

Note : Engines equipped with diesel particulate filters require the use of diesel fuel with 30 ppm sulfur maximum. There are no acceptable substitutes.

The viscosity of the fuel **must** be kept above 1.3 cSt at 40°C [104°F] to provide adequate pumping and lubricating characteristics to fuel system components.

The following chart lists acceptable substitute fuels for this engine.

Acceptable Substitute Fuels

Numb er 1D Diesel (1) (2) (3)	Numb er 2D Diesel (3)	Numb er 1K Keros ene	Jet-A	Jet- A1	JP-5	JP-8	Jet-B	JP-4	CITE
A	OK	Not OK	A	A	A	A	Not OK	Not OK	Not OK

Acceptable Substitute Fuels

Numb er 1D Diesel (1) (2) (3)	Numb er 2D Diesel (3)	Numb er 1K Keros ene	Jet-A	Jet- A1	JP-5	JP-8	Jet-B	JP-4	CITE
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1. An "A" means OK **only** if fuel lubricity is adequate. This means the BOCLE number is 3100 or greater as measured by ASTM specification D6078, Scuffing Load Ball On Cylinder Evaluator (SLBOCLE). Lubricity can also be measured by ASTM, specification D6079, ISO 12156, High Frequency Reciprocating Rig (HFRR) in which the fuel **must** have a wear scar diameter of 0.45 mm [0.02 in] or less.
2. Any adjustment to compensate for reduced performance with a fuel system using alternate fuel is **not** warrantable.
3. Winter blend fuels, such as found at commercial fuel-dispensing outlets, are combinations of number 1D and 2D diesel fuels and are acceptable.

Additional information for fuel recommendations and specifications can be found in Fluids for Cummins® Products Service Manual, Bulletin 5411406. See ordering information in the back of this manual.